Лабораторная работа 3. Фролов А.А. 2к, ИВТ-2

Изменение значения регистра

```
Z:\>debug
-r
AX=0000 BX=0000 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI NG NZ NA PO NC
0DAB:0100 C3                RET
-r AX
AX 0000  :3A7
-r
AX=03A7 BX=0000 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI NG NZ NA PO NC
0DAB:0100 C3                RET
-L
```

Сложение AX и BX

```
-e 100
0DAB:0100  01.01   D8.D8   00.
-e 100
0DAB:0100  01.     D8.     00.
-e 100
0DAB:0100  01.     D8.     00.
-e 101
0DAB:0101  D8.D8   00.
-r
AX=03A7 BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI NG NZ NA PO NC
0DAB:0100 01D8              ADD     AX,BX
-T
AX=0CD1 BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ AC PE NC
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:092A=00
```

Сложение AH и AL

```
-r ip
IP 0102  :100
-r
AX=0CD1 BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI PL NZ AC PE NC
0DAB:0100 01D8              ADD     AX,BX
-t
AX=15FB BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ NA PO NC
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:092A=00
```

Байтовая арифметика

```
-e 100
0DAB:0100  01.00   D8.C4   00.
-r
AX=15FB BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ NA PO NC
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:092A=00
-r IP
IP 0102  :100
-r
AX=15FB BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI PL NZ NA PO NC
0DAB:0100 00C4              ADD     AH,AL
-r AX
AX 15FB  :102
-r
AX=0102 BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI PL NZ NA PO NC
0DAB:0100 00C4              ADD     AH,AL
-t
AX=0302 BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ NA PE NC
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:092A=00
```

Вычитание AX и BX

```
AX=15FB BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ NA PO NC
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:092A=0
-e 100
0DAB:0100  01.29   D8.
-r ip
IP 0102  :100
-r
AX=15FB BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI PL NZ NA PO NC
0DAB:0100 29D8              SUB     AX,BX
-t
AX=0CD1 BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ NA PE NC
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:092A=0
-r ip
IP 0102  :100
-t
AX=03A7 BX=092A CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ AC PO NC
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:092A=0
-r ax
AX 03A7  :0
-r bx
BX 092A  :1
-r ip
IP 0102  :100
-t
AX=FFFF BX=0001 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI NG NZ AC PE CY
0DAB:0102 0000              ADD     [BX+SI],AL
```

Умножение AX на BX

```
-r
AX=0000 BX=0000 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI NG NZ NA PO NC
0DAB:0100 C3                RET
-r ax
AX 0000  :7c4b
-r bx
BX 0000  :100
-e 100
0DAB:0100  C3.E3   00.F7   00.
-r
AX=7C4B BX=0100 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI NG NZ NA PO NC
0DAB:0100 E3F7              JCXZ    00F9
-e 100
0DAB:0100  E3.F7   F7.E3   00.
-r
AX=7C4B BX=0100 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI NG NZ NA PO NC
0DAB:0100 F7E3              MUL     BX
-t
AX=4B00 BX=0100 CX=0000 DX=007C SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 OV UP EI NG NZ NA PO CY
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:0100=F7
```

Деление AX и BX

```
-e 100
0DAB:0100  F3.F7   F7.F3   00.
-r
AX=4B00 BX=0100 CX=0000 DX=007C SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 OV UP EI NG NZ NA PO CY
0DAB:0100 F7F3              DIV     BX
-r ax
AX 4B00  :4B12
-r
AX=4B12 BX=0100 CX=0000 DX=007C SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 OV UP EI NG NZ NA PO CY
0DAB:0100 F7F3              DIV     BX
-t
AX=7C4B BX=0100 CX=0000 DX=0012 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ NA PO CY
0DAB:0102 0000              ADD     [BX+SI],AL                      DS:0100=F7
```

Результат вычислений выражения, вариант 11

```
DOSBox-X 2026.01.02: DEBUG - 3000 cycles/ms
Главное меню  ЦП  Видео  Звук  DOS  Диск  Захват  Отладка  Справка
-e 100
0DAB:0100  F7.29   E3.d8   00.
-r
AX=68B0 BX=3181 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI PL NZ NA PE NC
0DAB:0100 29D8              SUB     AX,BX
-t
AX=372F BX=3181 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL NZ AC PO NC
0DAB:0102 0000              ADD     [BX+SI],AL                    DS:3181=00
-r bx
BX 3181  :11
-r ip
IP 0102  :100
-e100
0DAB:0100  29.F7   D8.F3   00.
-r
AX=372F BX=0011 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0100 NV UP EI PL NZ AC PO NC
0DAB:0100 F7F3              DIV     BX
-t
AX=033F BX=0011 CX=0000 DX=0000 SP=FFFE BP=0000 SI=0000 DI=0000
DS=0DAB ES=0DAB SS=0DAB CS=0DAB IP=0102 NV UP EI PL ZR NA PE NC
0DAB:0102 0000              ADD     [BX+SI],AL                    DS:0011=07
-
```

Записан в регистре AX = 33F